

**The University of Hong Kong
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE**

COMP7103 DATA MINING

Date: 15th Dec 2020

Time: 2:30pm – 4:30pm

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidate's responsibility to ensure that his/her calculator operates satisfactorily, and candidate must record the name and type of the calculator used at the beginning of their answer script.

IMPORTANT NOTE: Please be reminded that you have agreed to observe academic integrity. In particular, your submission with this exam is solely your own and you have not received any unauthorized assistance in this exam including unauthorized input from the Internet.

Answer **ALL** 4 questions.

1. Short questions (24%)

- (a) (6%) Explain the similarities and differences between regression analysis and classification analysis.
- (b) (6%) Explain the terms *supervised learning* and *unsupervised learning*.
- (c) (6%) Briefly describe a scenario in which support vector machine (SVM) is likely to perform better (in terms of classification accuracy) than decision tree classifiers (DT). You may describe an example dataset if that would help you explain your argument.
- (d) (6%) Briefly explain how the FP-growth algorithm could be more efficient than the Apriori algorithm in frequent itemset mining.

2. (25%) Consider the following dataset D of labeled objects (Table 1). In the table, each row shows an object id, two numerical attribute values (a_1 and a_2), and a class label.

Object	a_1	a_2	Class
X_1	10	20	C_1
X_2	15	10	C_2
X_3	15	15	C_3
X_4	15	10	C_1
X_5	15	20	C_1
X_6	20	10	C_2

Table 1: Dataset D

- (a) (3%) Perform min-max normalization on the given dataset.
- (b) (4%) Based on (the unnormalized) D , determine the best binary split if a decision tree is to be built by splitting D using attribute a_1 . Use *classification error* as the impurity measure. Show your calculation.
- (c) (4%) Based on (the unnormalized) D , build a decision tree by splitting the dataset using attribute a_2 only, assuming a three-way split.
- (d) (4%) Another decision tree is built as shown in Figure 1. Calculate the gain ratio of this split.

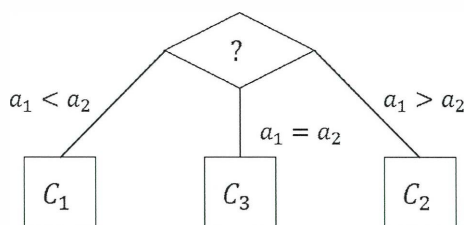


Figure 1: Classification tree

- (e) (6%) Calculate the accuracy, precision, and recall of class C_2 if the decision tree shown in Figure 1 is evaluated with the dataset in Table 1.
- (f) (4%) Give an example of data preprocessing operation that allows the decision tree shown in Figure 1 to be built.

3. (29%) Consider the market basket transaction dataset shown in Table 2.

Transaction ID	Items Bought
T_1	$\{a, b\}$
T_2	$\{a, b, d\}$
T_3	$\{a, b, d\}$
T_4	$\{a, c, d\}$
T_5	$\{a, b, c, d\}$
T_6	$\{b, d\}$

Table 2: Market basket transactions

- (a) (4%) Calculate the Simple Matching Coefficients and the Jaccard coefficients between Transactions T_1 and T_6 .
- (b) (6%) Give the support counts of all possible itemsets in Table 2. Then, identify all *closed* 2-itemsets.
- (c) (5%) If you are told that itemset $P = \{a, b, d\}$ is a maximal frequent itemset, is it possible that there are some other maximal frequent 3-itemsets? Explain your answer.
- (d) (6%) Suppose the transactions shown in Table 2 are the transactions conducted by a single customer in chronological order (T_1 first and T_6 last). A transaction sequence of length 6 is derived. Does this customer sequence contain $\langle \{c\}, \{d\}, \{c\}, \{d\} \rangle$? Explain your answer. Also, give one valid 8-sequence that is contained in the customer sequence.
- (e) (5%) Consider the Candidate Generation step of the Generalized Sequential Pattern (GSP) algorithm:
 “Merge two frequent $(k-1)$ -sequences w_1 and w_2 to produce a candidate k -sequence if the subsequence obtained by removing the **first** item in w_1 is the same as the subsequence obtained by removing the **last** item in w_2 .”
 Given that $\langle \{a\}, \{b\} \rangle$ and $\langle \{a\}, \{d\} \rangle$ are frequent 2-sequences, explain why the step does not consider merging $\langle \{a\}, \{b\} \rangle$ and $\langle \{a\}, \{d\} \rangle$ to produce the candidate $\langle \{a\}, \{b\}, \{d\} \rangle$.
- (f) (3%) If the Candidate Generation phase of the GSP algorithm is modified to the following:
 “Merge two frequent $(k-1)$ -sequences w_1 and w_2 to produce a candidate k -sequence if the subsequence obtained by removing the **last** item in w_1 is the same as the subsequence obtained by removing the **last** item in w_2 .”
 Given two frequent 2-sequences $\langle \{a\}, \{b\} \rangle$ and $\langle \{a\}, \{c\} \rangle$, what should be produced as candidates?

4. (22%) Consider the data objects listed in Table 3. Each object is associated with two numerical attributes x and y .

Object	x	y
O_1	0	3
O_2	3	6
O_3	9	15
O_4	21	36

Table 3: Dataset for cluster analysis

- (a) (4%) Construct the proximity matrix of the 4 objects using the *supremum distance* as the distance measure.
- (b) (6%) Perform k-means clustering with $k = 2$. Use Objects O_1 and O_4 as the initial centroids. Use the *supremum distance* as the distance measure.
- (c) (6%) Figure 2 shows a dendrogram built from the 4 data objects using the *supremum distance* as the distance measure. Is the dendrogram built using *single link* or *complete link* to determine inter-cluster similarity? Explain your answer. Complete the dendrogram by finding the missing proximity value indicated by ‘?’ in the dendrogram.

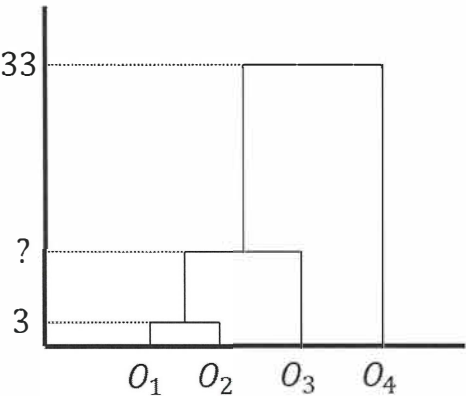


Figure 2: Dendrogram of 4 data objects

- (d) (6%) Construct 2 clusters using the dendrogram shown in Figure 2. Then, calculate the Silhouette Coefficient of each of the data objects using the *supremum distance* as the distance measure.

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